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Project Phase 2 - Data Collection -

CIS 517: Data Mining and Data Warehousing (First Semester)

Project Title: Predicting High-Risk Traffic Accident Locations Using Data Mining

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1. PROJECT OVERVIEW:

The project titled **"Predicting High-Risk Traffic Accident Locations Using Data Mining"** aims to leverage advanced data mining techniques to identify and predict areas with a high likelihood of traffic accidents. By analyzing historical accident data alongside environmental, temporal, and infrastructural factors, the project seeks to uncover hidden patterns that contribute to accident severity and frequency. The ultimate goal is to support traffic authorities in making informed, data-driven decisions to enhance road safety and reduce future incidents.

The selected dataset directly supports the project's objective of predicting high-risk traffic accident locations. Its rich set of attributes—including time, location, weather conditions, road infrastructure, and accident severity—enables comprehensive analysis of the factors contributing to traffic incidents. These variables are essential for building predictive models that can identify patterns and correlations, ultimately helping traffic authorities make data-driven decisions to improve road safety.

2. PHASE [2] INTRODUCTION:

This phase centers on the **collection of data** essential for conducting meaningful analysis in support of the project's objectives. The primary goal is to identify and obtain a dataset that is not only relevant and reliable but also sufficiently comprehensive to enable robust data mining techniques. The selected dataset must align with the analytical goals of the project and serve as a solid foundation for subsequent phases, including data preparation, modeling, and evaluation.

3. DATA SOURCES:

The dataset aggregates information from:

- **Dataset Name:** US Accidents (March 2023 500K and another datasets from GitHub from same resources if needed more datasets)
- **Source:** Kaggle and GitHub
- **Size:** Approximately 500,000 records with 46 attributes
- **Format:** Excel (converted from CSV)
- **Reason for Selection:**
  - Includes detailed accident records from multiple states
  - Covers key features such as location, time, weather, road conditions, and severity.
  - Widely used in research, ensuring reliability and relevance.

4. ACCESS AND RELEVANCE

The dataset is freely available for academic use in CSV format, ensuring compatibility with popular data mining tools such as Python, R, and Weka. Its comprehensive coverage of accident-related variables makes it highly relevant for classification, clustering, and frequent pattern mining tasks.

5. DATA AVAILABILITY CHECK

To ensure that the dataset fully supports our project objectives, we compared each required factor (weather conditions, road infrastructure, traffic congestion, time of day, and geographic location) with the actual columns in the **US_Accidents dataset**.

The following table summarizes the comparison, confirming that all required data fields are present.

Table 1 Data Availability Comparison

Requirement	Needed Columns	Availability
Weather / Environmental Conditions	Temperature(F), Humidity(%), Weather Condition, Precipitation(in), Windspeed(mph), Visibility(mi)	Yes
Road Infrastructure Details	Street, Junction, Traffic Signal, Crossing, Bump, Roundabout, Stop, Station	Yes
Traffic Congestion	Distance(mi)	Yes
Time of Day	Start_Time, End_Time, Sunrise Sunset, Civil Twilight, Nautical_Twilight, Astronomical_Twilight	Yes
Location / Geographic	City, County, State, Zipcode, Start_Lat, Start_Lng	Yes

Table 2. Data Availability Comparison

6. RELIABILITY AND SUFFICIENCY

The dataset has been used in multiple research studies and is considered a benchmark for traffic accident analysis. Its large volume and multi-year coverage ensure statistical significance, making it suitable for both spatial and temporal analysis.

In this phase, we will use a sampled version of the dataset titled **US_Accidents_March23_sampled_500k**, which contains **500,000 records**. This version is selected to balance computational efficiency with analytical depth, allowing us to perform meaningful data mining while maintaining manageable processing time and resource usage.

Table 3 Overview of Dataset Attributes

Column Name	Description	Example Value
ID	Unique identifier for each accident	A-1
Source	Data source provider	MapQuest
TMC	Traffic Message Channel code	201.0
Severity	Accident severity (1–4)	3
Start_Time	Start date and time of accident	2016-02-08 05:46
End_Time	End date and time of accident	2016-02-08 11:00
Start_Lat	Latitude of start location	39.865147
Start_Lng	Longitude of start location	-84.058723
End_Lat	Latitude of end location	39.865147
End_Lng	Longitude of end location	-84.058723
Distance(mi)	Distance affected in miles	0.01
Description	Text description of the accident	Right lane blocked
Number	Street number	2584.0
Street	Street name	Brice Rd
Side	Road side (L/R)	R
City	City name	Dayton
County	County name	Montgomery
State	State abbreviation	OH
Zipcode	Postal code	45424
Country	Country code	US
Timezone	Time zone of location	US/Eastern
Airport_Code	Nearest airport code	KFFO
Weather_Timestamp	Time of weather observation	2016-02-08 05:58
Temperature(F)	Temperature in Fahrenheit	36.9
Wind_Chill(F)	Wind chill in Fahrenheit	33.3
Humidity(%)	Humidity percentage	91.0
Pressure(in)	Atmospheric pressure in inches	29.68
Visibility(mi)	Visibility in miles	10.0
Wind_Direction	Wind direction	Calm
Wind_Speed(mph)	Wind speed in mph	3.5
Precipitation(in)	Precipitation in inches	0.02
Weather_Condition	Weather condition description	Light Rain
Amenity	Nearby amenity (True/False)	False
Bump	Bump present (True/False)	False
Crossing	Crossing present (True/False)	False

Give_Way	Give way present (True/False)	False
Junction	Junction present (True/False)	False
No_Exit	No exit present (True/False)	False
Railway	Railway present (True/False)	False
Roundabout	Roundabout present (True/False)	False
Station	Station present (True/False)	False
Stop	Stop sign present (True/False)	False
Traffic_Calming	Traffic calming present	False
Traffic_Signal	Traffic signal present	False
Turning_Loop	Turning loop present	False
Sunrise_Sunset	Day/Night indicator	Night
Civil_Twilight	Civil twilight indicator	Night
Nautical_Twilight	Nautical twilight indicator	Night
Astronomical_Twilight	Astronomical twilight indicator	Night

NEXT PHASE: DATA PREPARATION:

In the next phase, we will:

- **Data Cleaning:** Handle missing values, remove duplicates, and correct inconsistencies.
- **Data Transformation:** Normalize numeric features, encode categorical variables, and engineer new features (e.g., accident duration, day/night).
- **Data Splitting:** Divide the dataset into training (70%) and testing (30%) sets for predictive modeling.